Answers and talk notes

Math is Explaining and Sharing

What your student is learning

Tonight we do not just compute — we recommend. We compare two garden bed designs by their volume and then say, in words, which one we would build and why.

Answers

Ask your student to explain before you look.

1. **D.** cubic feet. The unit is cubic feet. Volume counts cubes that fill the space inside the bed.
2. **A.** 20 cubic feet. 5 × 2 × 2 = 20 cubic feet.
3. **C.** The volumes are equal, so you need another reason to choose. The volumes are equal, so you need another reason to choose.
4. **B.** Bed A, because 18 cubic feet is more than 16 cubic feet. Bed A, because 18 cubic feet is more than 16 cubic feet. Bed A: 3 × 3 × 2 = 18. Bed B: 4 × 2 × 2 = 16.
5. **D.** A number you computed and a reason it settles the question. A number you computed and a reason it settles the question.

Talk about it

Do the two boxes hold the same amount? What other reason could help you choose?

Home mission

A different situation: use 12 blocks, dice, or ice cubes. Build a box 3 long, 2 wide, and 2 tall. Then build one that is 4 long, 3 wide, and 1 tall. Do they hold the same amount? Which would your student recommend?

When they are stuck, ask

- What number did you find, and what question does it answer?
- Is soil filling a space, covering a surface, or going along a line?
- Can you say your recommendation in one sentence that includes the number?

Try not to say

- The units do not matter as long as the number is right.
- Just pick the bigger one, that is always better.
- You do not have to explain, the answer speaks for itself.